Ex: 9
Date: 20/8/26

Implementation of Subnetting in CISCO packet Tracer simulator

a.) Design Multiple Subnets with a suitable Number of the hosts.

AIM :-

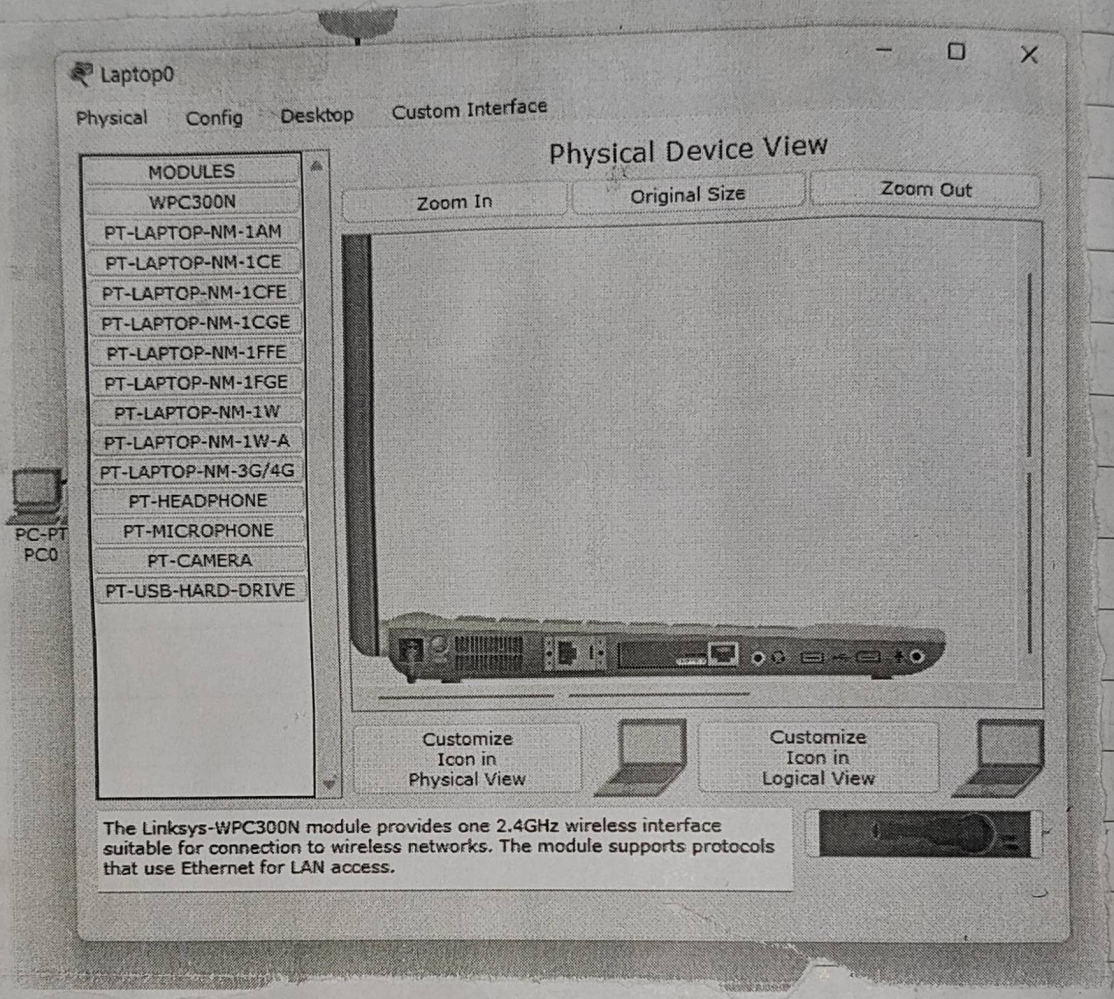
To implement Subnetting by individual a large network into multiple Smaller subnetworks according to the required number of hosts and to study the allocation of IP addresses in Cisco Packet Tracer.

Steps :-

1.) Open Cisco packet Tracer and Create a new project.
2.) Design the network topology by placing the required network devices such as routers, switches and PC's.
3.) Select a suitable network address for the experiment
4.) Calculate the number of Subnets required.
5.) Determine the number of hosts needed in each subnet
6.) Calculate the subnet mask using subnetting techniques.
7.) Divide the network into multiple subnets.
8.) Assign a seperate networks address to each subnet.
9.) Verify the subnet Calculations and host ranges.
10.) Save the network topology.

RESULT :-

Thus, the multiple subnets were successfully created according to the required number of hosts and the network address was efficiently divided into smaller subnets.


Laptop0
Physical
Config
Desktop
Custom Interface
Physical Device View
MODULES
WPC300N
PT-LAPTOP-NM-1AM
PT-LAPTOP-NM-1CE
PT-LAPTOP-NM-1CFE
PT-LAPTOP-NM-1CGE
PT-LAPTOP-NM-1FFE
PT-LAPTOP-NM-1FGE
PT-LAPTOP-NM-1W
PT-LAPTOP-NM-1W-A
PT-LAPTOP-NM-3G/4G
PT-HEADPHONE
PT-MICROPHONE
PT-CAMERA
PT-USB-HARD-DRIVE
Zoom In
Original Size
Zoom Out
PC-PT
PC0
Customize
Icon in
Physical View
Customize
Icon in
Logical View
The Linksys-WPC300N module provides one 2.4GHz wireless interface suitable for connection to wireless networks. The module supports protocols that use Ethernet for LAN access.

Ex: 9b)

Assign Static IP Address across all subnet and Connect the subnet through a Router

AIM:-

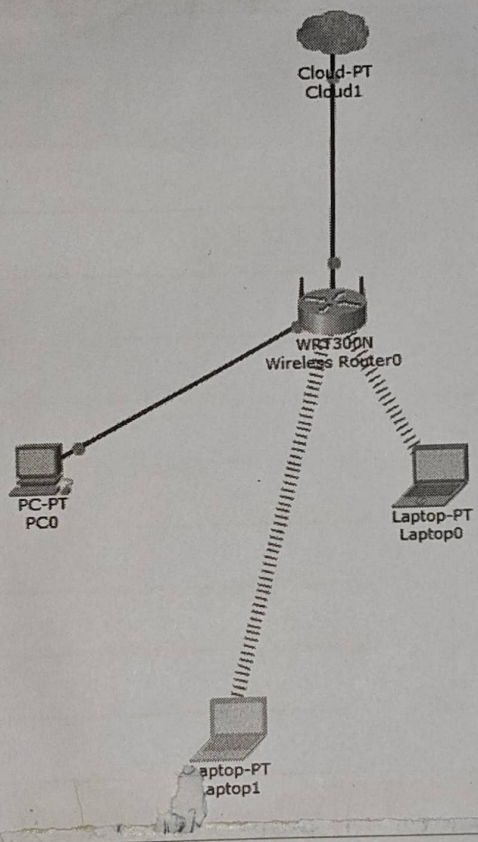
To assign static IP address to all devices in different subnets and establish communication between the subnets through a router.

Steps :-

1.) Place a router in the network topology.
2.) Connect the router to the switches using appropriate Cables.
3.) Configure the router interfaces.
4.) Assign and IP address and subnet mask to each router interfaces.
5.) Configure the switches
6.) Assign static IP address to all computers in every subnet.
7.) Configure the subnet mask for each devices.
8.) Assign the appropriate default gateway to each computer
9.) Verify the IP configuration of all devices.
10.) Save the router configurations.
11.) Check the network topology for configuration errors.
12.) Verify that all devices are properly connected.

RESULT :-

Thus, static IP address were successfully assigned to all devices and the communication between different subnets was establish through the router.


Cloud-PT
Cloud1
WRT300N
Wireless Router0
PC-PT
PC0
Laptop-PT
Laptop0
aptop-PT
aptop1

Ex: 9c

Simulate Packet Transmission Across the Subnets and observe the result

(i) When Subnets are connected through a router

AIM:-

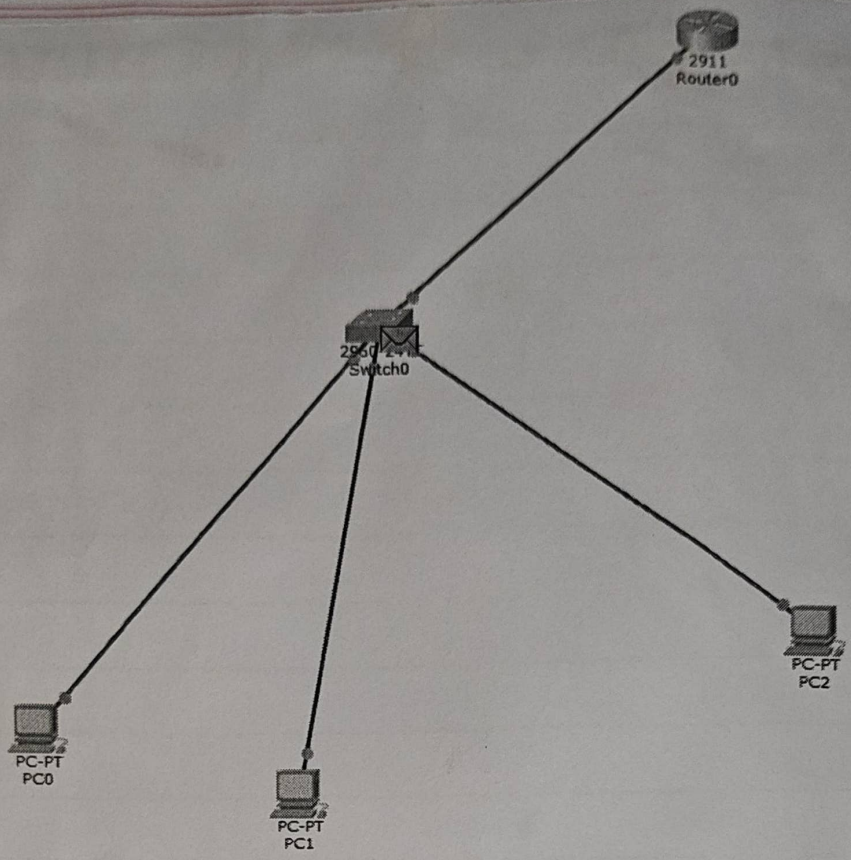
To analyze a packet transmission between different subnets when a router is used for communication

Steps:-

1.) Open Simulation Mode in Cisco Packet Tracer
2.) Select a Source devices from one ~~source~~ Subnet
3.) Select a destination devices from another subnet
4.) Generate a packet using the sample PDU Tool.
5.) Observe the movement of the packet through the network
6.) Analyze how the router receive the packet.
7.) Observe the forwarding process performed by the router.
8.) Verify the packet transmission between different subnets.
9.) Use the ping command to test connectivity.
10.) Observe the successful delivery of packets.
11.) Record the simulation result.
12.) Analyze the routing process.

RESULT:-

Thus, packets were successfully transmitted between different subnets through the router. The router received the packet from the source subnet, processed them and forward them to the destination subnet. successful ping response confirmed proper communication between all network devices.




```
        --- System Configuration Dialog ---

Continue with configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitethernet 0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state
to up
exit
```

ii.) When Subnets are not connected through a router

AIM :-

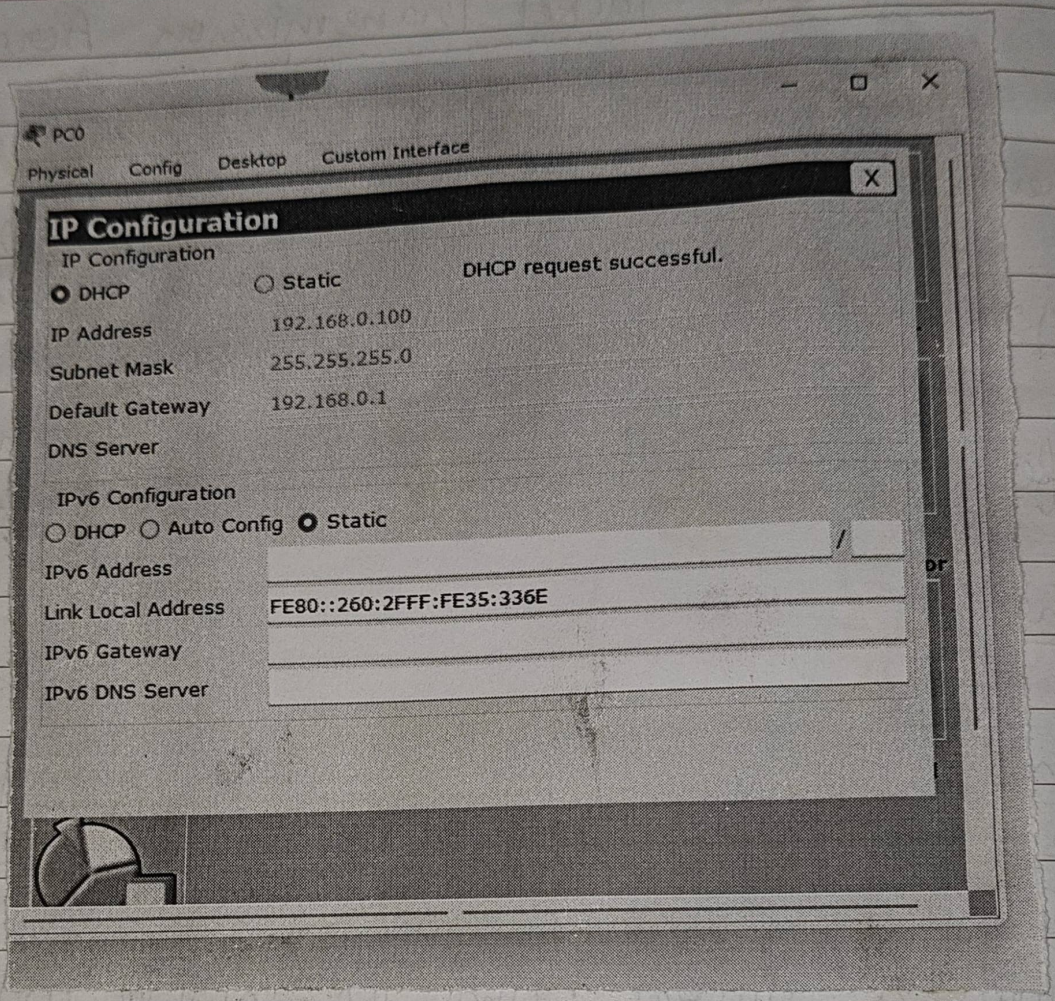
To analyze packet transmission between different subnets when no router is available.

Steps :-

1.) Remove the router from the network topology
2.) Open simulation Mode.
3.) Select devices belonging to diffrent Subnets.
4.) Generate packets using the simple PDU tools.
5.) Observe the packets transmission.
6.) Perform a ping test between devices located in different subnets.
7.) Analyze the packet flow.
8.) Observe the simulation events.
9.) Identify the point at which packet transmission fails.
10.) Compare the result with the network that uses a router.
11.) Record the Observation.
12.) Analyze the importance of a router in inter-network communication.

RESULT :-

Thus, Communication between devices located in different subnets failed when a router was not used. The packets could not reach the destination because there was no routing devices to forward them. This experiment demonstrated the important of routers in enabling communication between different subnet works


PC0
Physical
Config
Desktop
Custom Interface
IP Configuration
IP Configuration
DHCP
Static
DHCP request successful.
IP Address
192.168.0.100
Subnet Mask
255.255.255.0
Default Gateway
192.168.0.1
DNS Server
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
Link Local Address
FE80::260:2FFF:FE35:336E
IPv6 Gateway
IPv6 DNS Server